

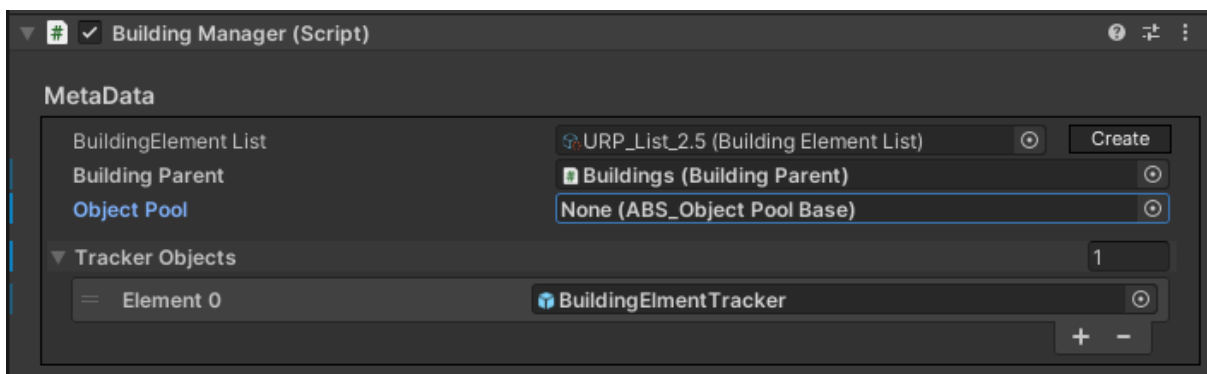
v1.4.0

The goal of this documentation is to give a short demo and some information about the new features of the version 1.4.0. This document does not explain everything in detail. For a detailed explanation read the documentation of the asset! Also note that not every change is included by this document, only the new features.

ObjectPool/Buffer

From now the BuildingManager can use ObjectPools. For that an object should be provided on the manager's interface which is inherited from a new abstract class

ABS_ObjectPoolBase The goal of the object pool feature is to increase the performance of the drag building and avoid the fps drop when a huge amount of elements is created at the same time.



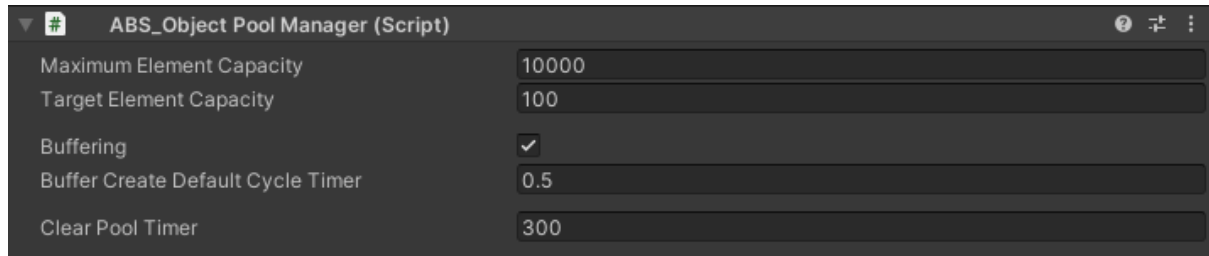
```
namespace REST.AdvancedBuildSystem.src
{
    public abstract class ABS_ObjectPoolBase : MonoBehaviour
    {
        public abstract BuildingElement Get(BuildingElement p_Element);
        public abstract void GiveBack(BuildingElement p_Element);
        public abstract void Release(BuildingElement p_Element);
    }
}
```

The ObjectPool should implement the following 3 functions

- Get
 - An element prefab is provided as a parameter.
 - A BuildingElement instance (GameObject instantiated from the p_Element) should be returned.
- GiveBack
 - A BuildingElement instance is provided as a parameter.
 - This element should be placed back to the pool.

- Release
 - A BuildingElement instance is provided as a parameter.
 - The Object pool should release the reference of the given element.

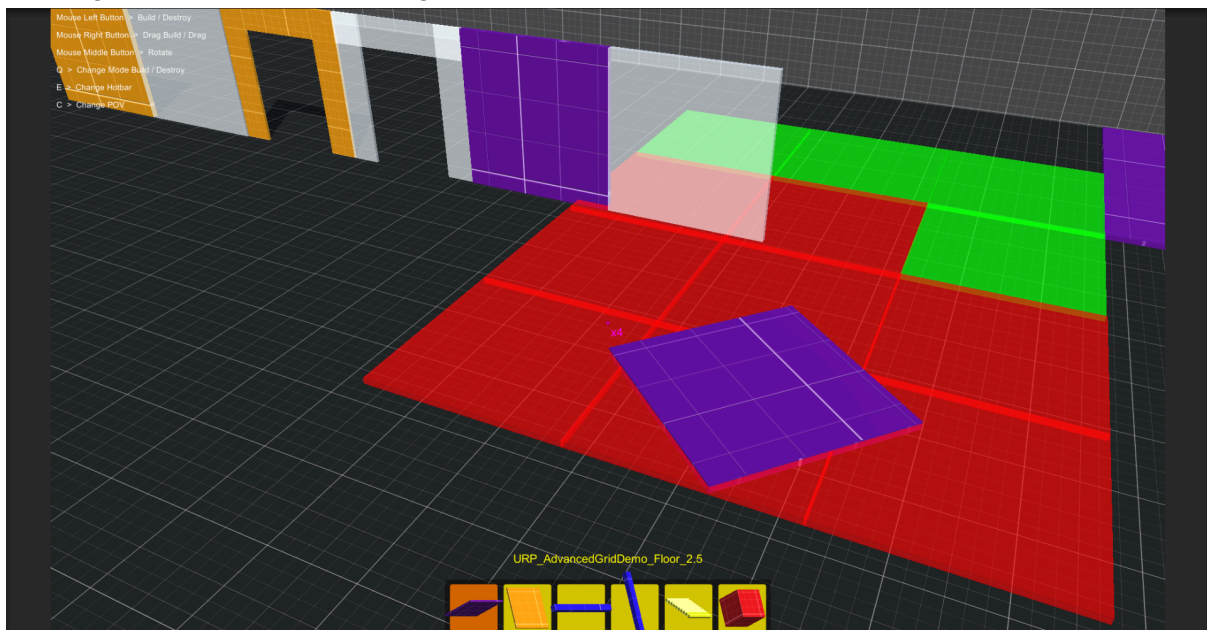
The AdvancedBuildingSystem has its own implementation, the ABS_ObjectPoolManager. This is the first version of the ABS own Object pool. The ABS_ObjectPoolManager has a buffering mechanism next to the pool functionalities and a clearing feature for memory saving purposes.



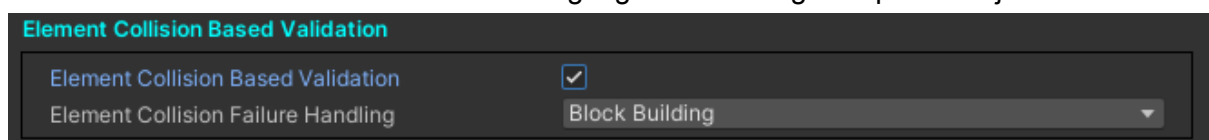
Validation changes

BuildingElement collision

The building algorithms automatically checked the collision of building elements from other Buildings and denied the building on that position.

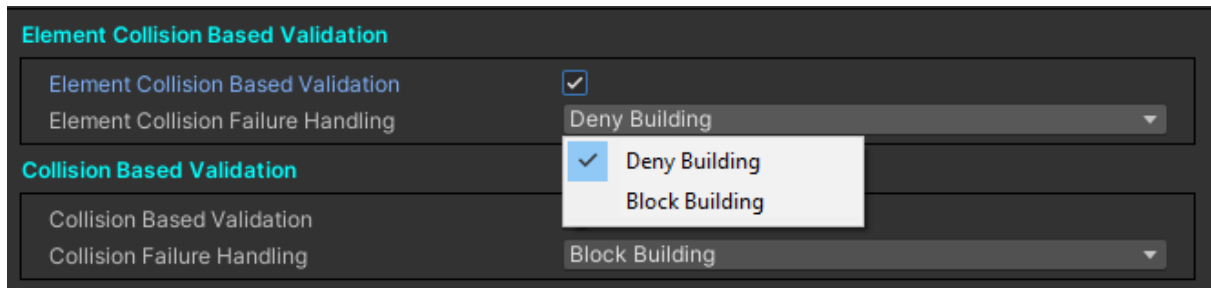


From now it can be turned off on the building algorithm settings scriptable objects.

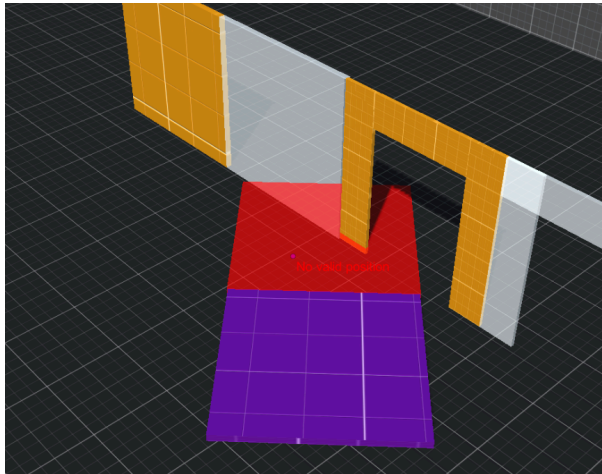


Failure Handling

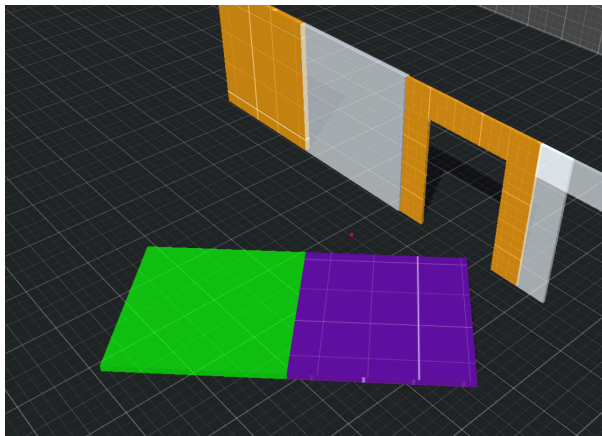
Some validation has a new logic where that can be decided if the building of an element to that position is denied or blocked.



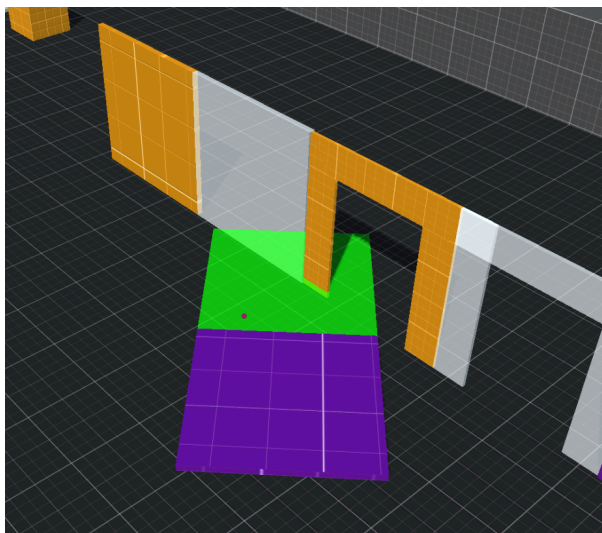
Block: The element can be placed there but it will be blocked.



Deny: The element can't be placed there and the algorithm will try to find another position.



The feature is disabled so the element can be placed there and its placement is allowed.



Building Range Limits

From now the Buildings can have range limitations that are calculated from the Building's zero localPosition (0,0,0). If the limit is enabled, the range check will be part of the validation process. If the element is over or under the range limit its placement will be blocked.

The screenshot shows the 'Advanced Grid Building (Script)' settings window. At the top, there's a 'Building Position Modifier' section with X, Y, and Z coordinates set to 0, and a 'Refresh Modifier' button. Below this, the 'Maximum Element Count' is set to 1000. A red box highlights the following settings:

- Upper Range Limit Enabled:** ☒
- Upper Range Limit:** 1000
- Under Range Limit Enabled:** ☒
- Under Range Limit:** -1000
- Side Rangenet Limit Enabled:** ☒
- Side Rangenet Limit:** 1000

Below the red box, there are sections for 'Cache' (with 'Clear Cache' and 'Check Cache Size' buttons, and 'Use Cache' checked), 'Modifications' (with 'Find All And Replace' button), and 'Persistence' (with 'Save To File' and 'Save To Prefab' buttons).

The limitations can be set on the BuildingParent too so the limitations will be applied on every new Building.

The screenshot shows the 'Building Parent (Script)' settings window. It has fields for 'Global BasicGrid Parent' (set to 'BasicGridBuildingParent (Basic Grid Building)'), 'Global Free Parent' (set to 'FreeBuildingParent (Free Building)'), 'New Buildings Default Name' (set to 'Building'), and 'Enable Cache' (checked). Below these, there are four fields for maximum element counts, all set to 1000:

- Maximum Element For FreeBuilding: 1000
- Maximum Element For BasicGridBuilding: 1000
- Maximum Element For AdvancedGridBuilding: 1000
- Maximum Element For SnapPointBasedBuidling: 1000

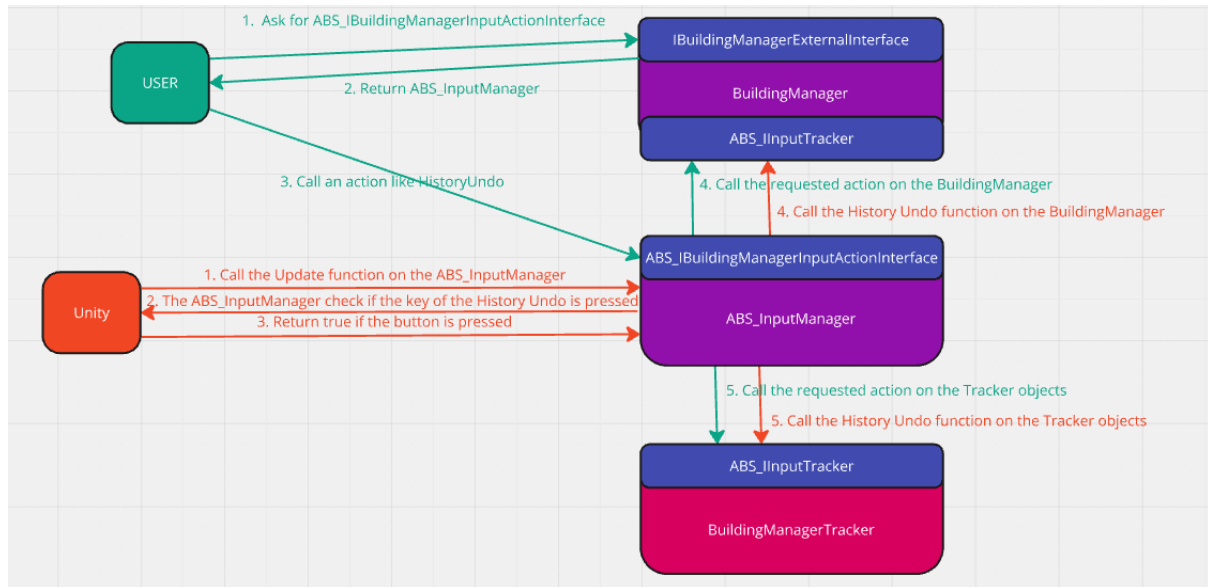
A red box highlights the following settings:

- Upper Range Limit Enabled:** ☐
- Under Range Limit Enabled:** ☐
- Side Rangenet Limit Enabled:** ☐

At the bottom, there are two buttons: 'Init Free Building Parent' and 'Init BasicGrid Building Parent'.

New Input System

A new component has been implemented to separate the input handling from the BuildingManager. This new InputManager is managed by the BuildingManager so the developers should not care about that. The new system can handle the legacy key based inputs from Unity and also it can be managed by the developers via messages like OnDragBuildingPressed or OnDragBuildingReleased.



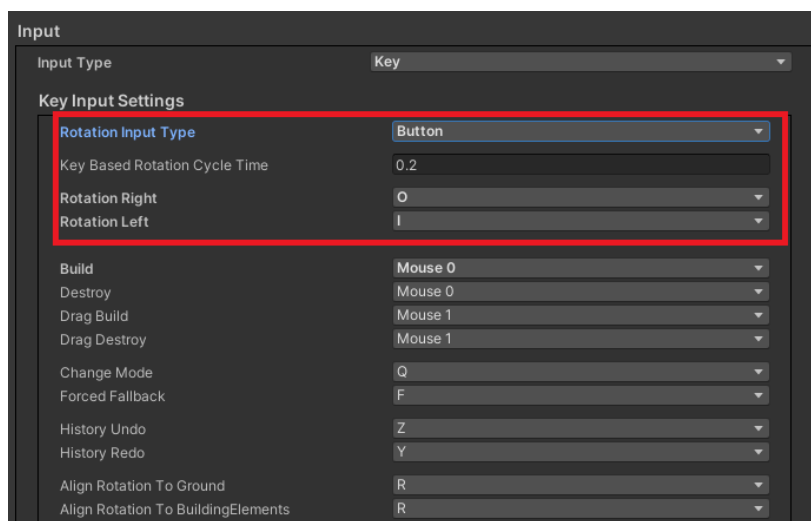
You can trigger those message functions through a new interface, the ABS_BuildingManagerInputActionInterface. You can get the input interface from the IBuildingManagerExternalInterface of the BuildingManager. With the IBuildingManagerExternalInterface interface the settings with the keybindings can be changed during runtime anytime.

```

public interface IBuildingManagerExternalInterface
{
    //+++++
    // Input
    //+++++

    /// <summary>
    /// Getter for the Input Interface of the BuildingManager
    /// </summary>
    /// <returns> The Input Interface of The BuildingManager.</returns>
    public ABS_BuildingManagerInputActionInterface GetInputInterface();
}
    
```

From now the rotation can be managed by buttons not just with the mouse wheel:



From now the BuildingManagerTracker will get the InputManagers inputs as function calls to give the possibility to react for the same input with the BuildingManager. For that a new interface is introduced, the ABS_IInputTracker.

Note that the BuildingManagerTracker is implementing the new interface so you can implement those functions and react to them by inheriting from the tracker.

```
9
10 namespace REST.AdvancedBuildSystem.src
11 {
12     public interface ABS_IInputTracker
13     {
14         //Change BuildingMode
15         public void ChangeModePressed();
16
17         //BuildingProcess
18         public void ForcedFallbackPressed();
19         public void SimpleBuildingPressed();
20         public void DragBuildingPressed();
21         public void DragBuildingReleased();
22
23         //Rotation
24         6 references | Added by resb014@gmail.com on Sunday, February 16, 2025
25         public void MouseWheelChanged(in float p_Value);
26         6 references | - Plastic SCM
27         public void RotationButtonRightPressed();
28         6 references | Added by resb014@gmail.com on Sunday, February 16, 2025
29         public void RotationButtonRightHold();
30         6 references | Added by resb014@gmail.com on Sunday, February 16, 2025
31         public void RotationButtonRightReleased();
32         6 references | Added by resb014@gmail.com on Sunday, February 16, 2025
33         public void RotationButtonLeftPressed();
34         6 references | Added by resb014@gmail.com on Sunday, February 16, 2025
35         public void RotationButtonLeftHold();
36         6 references | Added by resb014@gmail.com on Sunday, February 16, 2025
37         public void RotationButtonLeftReleased();
38
39         //RotationAligment
40         6 references | Added by resb014@gmail.com on Saturday, February 15, 2025
41         public void AlignRotationToGroundPressed();
42         6 references | Added by resb014@gmail.com on Sunday, February 16, 2025
43         public void AlignRotationToGroundReleased();
44         6 references | Added by resb014@gmail.com on Saturday, February 15, 2025
45         public void AlignRotationToBuildingElementsPressed();
46
47         //History
48         6 references | Added by resb014@gmail.com on Saturday, February 15, 2025
49         public void UndoPressed();
50         6 references | Added by resb014@gmail.com on Saturday, February 15, 2025
51         public void RedoPressed();
52
53         //Destroy
54         6 references | Added by resb014@gmail.com on Saturday, February 15, 2025
55         public void SimpleDestroyPressed();
56         6 references | Added by resb014@gmail.com on Saturday, February 15, 2025
57         public void SimpleDestroyHold();
58         6 references | Added by resb014@gmail.com on Saturday, February 15, 2025
59         public void SimpleDestroyReleased();
60         6 references | Added by resb014@gmail.com on Saturday, February 15, 2025
61         public void DragDestroyPressed();
62         6 references | Added by resb014@gmail.com on Saturday, February 15, 2025
63         public void DragDestroyReleased();
64     }
65 }
```